

Mrinal Paul

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OBJECTIVE

High-agency Computer Science undergraduate combining rigorous backend architecture, FastAPI/Python microservices, and relational database modeling with data fluency and automation tooling. Proven experience building production-grade, multi-tenant full-stack systems and scaling user communities.

SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, Java, C, HTML5, CSS3
- **Development & Database:** HTML5, CSS3, JavaScript, React.js, Tailwind CSS, MySQL, Workflow Automation (Google Apps Script / Scripts)
- **Databases & Cloud:** MySQL, Relational/Document Data Modeling, AWS Cloud Fundamentals, Git, Docker
- **Machine Learning & AI:** TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib
- **Concepts:** DSA (Java), OOPs, Machine Learning, Artificial Intelligence, Deep Learning, Data Science, Programmer Analyst and BI Analyst pipelines.
- **Soft Skills:** Problem Solving, Team Collaboration, Rapid Learning, Leadership, Communication
- **Design & Collaboration Tools:** Figma, Canva, VS Code, Postman

EDUCATION

- **SRM Institute of Science and Technology KTR, Chennai** Expected 2027
Bachelor of Technology – B. Tech - CSE | CGPA: 9.09
- **St. Patrick's H.S. School, Asansol** 2022 – 2023
Higher Secondary Certificate Programs, Class 12th | Grade: 86.25%
- **St. Vincent's High and Technical School, Asansol** 2011 - 2021
Secondary Certificate Programs, Class 10th | Grade: 85.4%

RESEARCH WORK

Next-Gen Farming: Machine Learning–Driven Disruptive Technologies | Co-Author

Undergraduate Research Opportunity Program (UROP) | SRM Institute of Science and Technology

Publication Status: Published by ARCC Journal (Scopus Indexed); doi: 10.18805/IJARE.A-6504.

Tech Stack: Python, Scikit-learn, TensorFlow, Keras, NumPy, Pandas, Matplotlib

- Engineered an AI-driven precision agriculture framework, achieving an R^2 score of 0.9797 for crop yield prediction using a Random Forest Regressor on 19,698 historical records.
- Developed a Convolutional Neural Network (CNN) trained on 50,000 augmented images to classify 38 crop disease classes with 87.69% accuracy and an F1-Score of 0.8363.
- Implemented K-Means clustering for soil classification (Silhouette Score: 0.3647) and integrated all models into a Decision Intelligence Layer to generate actionable farming recommendations.

HealthTwin: Proactive Health Assistance (CKM Triad) | Ongoing Research

Tech Stack: Python, XGBoost, Random Forest, HistGradientBoosting (LightGBM), Streamlit, Pandas.

- Engineered a proactive, "Assistive AI" predictive engine integrating Cardiovascular, Kidney, and Metabolic (CKM Triad) health, trained on 100,000+ real-world clinical and lifestyle records from the CDC and NHANES.
- Developed an Ensemble Learning pipeline utilizing XGBoost for metabolic early warning, Random Forest for cardiac risk classification, and HistGradientBoosting for renal stage detection based on complex laboratory markers.
- Designed an interactive Streamlit dashboard featuring dynamic health scoring, visual risk gauges, and one-click automated PDF clinical reporting to support data-driven physician decision-making.

ACADEMIC SERVICE & PEER REVIEW

Official Peer Reviewer | IEEE Access | March 2026

Identifiers: ORCID: 0009-0006-8427-0172 | **Web of Science ResearcherID:** PNF-4111-2026

- Evaluated and validated foundational AI research for the manuscript: "VeriGen: An LLM-Augmented Framework for End-to-End Automation of Software Development Lifecycle."
- Critically evaluated software engineering manuscripts focused on LLM-augmented frameworks for automated **software development lifecycle (SDLC)**, code synthesis verification loops, and system reliability
- Contributed to the advancement of the software engineering research community by reviewing the empirical validity of Verifier-Optimizer loops designed to mitigate LLM hallucinations and structural inconsistencies.

PROJECT WORKS

1. Employee Management System (EMS)

Tech Stack: MySQL, PHP, Python, HTML5, CSS3, JavaScript

- Engineered a full-stack Employee Management System utilizing Python/FastAPI backend logic and MySQL relational databases to automate attendance, payroll, and role-based access control (RBAC).
- Modeled complex multi-tenant database schemas and optimized indexing to guarantee low-latency query execution and secure administrative workflows.

EXPERIENCE

1. Front-End Web Development Intern

AICTE–Edunet Foundation (In Collaboration with IBM SkillBuild Virtually)

Aug 2025 – Sept 2025

- Completed a 6-week internship focused on responsive web design and frontend development using HTML, CSS, and JavaScript.
- Built interactive UI components and improved site usability following IBM SkillBuild best practices.
- Enhanced understanding of modern frontend frameworks and web accessibility standards.

2. IIT Kharagpur

Research Internship

May 2026 – July 2026

- Applied AI/ML for nutrient automation, disease detection, and stress diagnosis in hydroponic systems.
- Optimized crop-specific nutrient requirements and formulated organic nutrient solutions for sustainable yield.
- Monitored crop growth using image processing and hyperspectral sensing for precision agriculture.
- Analyzed crop and environmental data to enhance hydroponic efficiency and food security.

LEADERSHIP

1. President & Campus Lead

GeeksforGeeks (Representing SRM Institute Science and Technology)

Apr 2025 – Present • Chennai, India

- Led a 40-member core team and built an active virtual developer community of 800+ students from scratch.
- Organized "IGNISIA 1.0" for 150+ attendees, focusing on practical software development using modern AI tools like Google Gemini and ChatGPT.
- Executed "JAVA-VERSE 2026", driving 450+ registrations for hands-on technical workshops covering Java Full-Stack, Spring Boot, and Microservices.

2. Campus Ambassador

Unstop | SRM Institute of Science and Technology

- Community Growth:** Spearheaded campus outreach, scaling a highly engaged student community.
- Strategic Partnerships:** Negotiated sponsorships and orchestrated collaborations with university clubs.
- Event Management:** Executed end-to-end coding challenges, hackathons, and workshops, connecting SRMIST engineering talent with industry opportunities.

3. Campus Ambassador

GirlScript Summer of Code (GSSoC) | Remote Internship

Sept 2024 – Feb 2025

- Ranked AIR-24 nationwide for onboarding 140+ open-source contributors across India.
- Conducted events, webinars, and outreach sessions promoting open-source development and collaboration.
- Received Letter of Recommendation and Top Performer Certificate.

ACHIEVEMENTS

- SRM Project EXPO 2026** – 2nd Position, Innovation Category
- EY Techathon 6.0** – Semi-Final Round
- RedBull Hackathon** (Internal) – Top 20 Teams
- GirlScript Summer of Code** – Ranked AIR-24 nationwide Campus Ambassador

CERTIFICATIONS

- Remote Sensing and Digital Image Analysis — ISRO–IIRS, Dehradun (Aug–Sep 2025)
- Google Cloud: Agentic AI Day & Gen AI Exchange '25 — Google (Sep 2025)
- Applied AI: Deep Learning, NLP, & Computer Vision 101 — Infosys Springboard 6.0 (Jun 2025)
- Image Processing Onramp — MathWorks (Mar 2025)
- Introduction to Databases — Meta (Apr 2025)
- Oracle Cloud Infrastructure 2024 — Oracle (Jan 2025)
- AWS Academy: Data Engineering & Cloud Foundations — Amazon Web Services (Feb 2024)
- Participant: Bharatiya Antariksh Hackathon & Adobe India Hackathon — ISRO & Adobe (Aug–Sep 2025)